Model Predictive Control Final

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***Abstract*—In this study, various controller designs were carried out based on the linearized continuous-time model of the Cessna Citation aircraft at an altitude of 5000 meters and a speed of 128.2 m/s. First, the discrete-time version of the system was obtained. Then, control performances were examined using classical continuous-time LQR, discrete-time LQR, input-constrained LQR, and various constrained/unconstrained Model Predictive Control (MPC) methods. For each controller, the initial condition was kept constant, and the control input and system responses were evaluated comparatively. In addition, the effects of input constraints, the role of MPC horizon length on system stability, and the simulation results were analyzed.**

***Index Terms*—Model Predictive Controller, LQR Controller, Constraint, Discrete Time, Cessna Citation, Continuous Time**

I. Introduction

In this study, the effects of classical LQR and Model Predictive Control (MPC)-based controllers on the linearized continuous-time model of the Cessna Citation aircraft are compared. While the LQR method provides a stable control approach, MPC enables the analysis of the impact of parameters such as control input constraints and prediction horizon on system stability. For both control methods, state variables such as altitude, pitch angle, pitch rate, and angle of attack are evaluated.

II. Materials and Methods

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A. Discrete-Time Model with Euler Approximation

Given the continuous-time linearized state-space model of the Cessna Citation aircraft, the discrete-time equivalent of the system was obtained using the Euler approximation method. The MATLAB function `getEulerSS(A,B,C,D,dT)` was utilized to compute the following discrete-time matrices with a sampling time of $T_s = 0.25$ seconds:

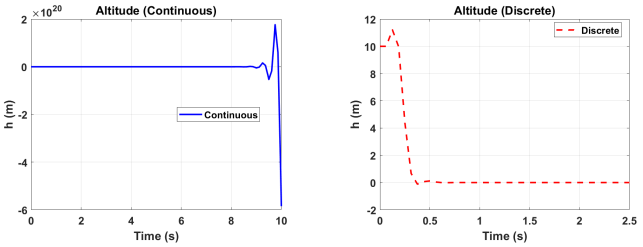
$$A_d = \begin{bmatrix} 0.6794 & 0 & 0.2450 & 0 \\ 0 & 1.0000 & 0.2500 & 0 \\ -1.3561 & 0 & 0.5409 & 0 \\ -32.0500 & 32.0500 & 0 & 1.0000 \end{bmatrix}$$

$$B_d = \begin{bmatrix} -0.0750 \\ 0 \\ -4.2500 \\ 0 \end{bmatrix} \quad C_d = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad D_d = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

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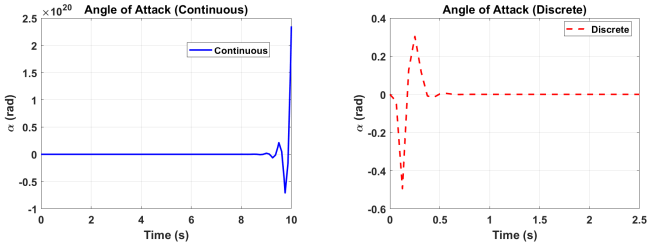
B. LQR Response Comparison

Below are the responses of the continuous-time and discrete-time systems under the LQR controller for each state and control input. As seen in the discrete-time plots, all variables exhibit an initial overshoot and undershoot—for example, an altitude peak of approximately 110 percent above the setpoint and an angle of attack trough near –0.5 rad—yet the system settles into the ±2 percent band within 0.3–0.4 s. The control input reaches up to ±0.6 rad to achieve this aggressive correction, while the steady-state value of every state variable remains zero. These results demonstrate the balanced performance afforded by the chosen LQR weighting matrices. In contrast, the continuous-time plots reveal that applying the same gains directly in continuous time leads the system toward instability.



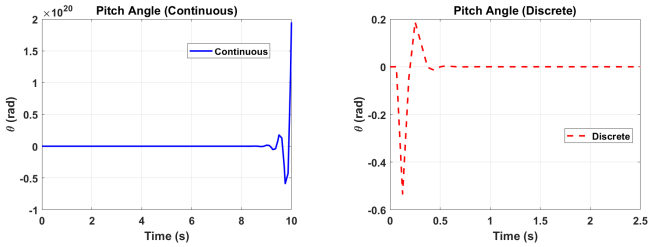
(a) Altitude (Continuous) (b) Altitude (Discrete)

Fig. 1: Altitude response under LQR control.



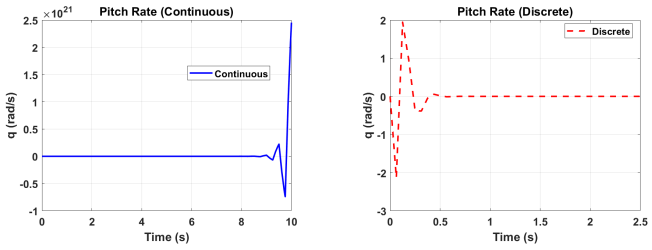
(a) Angle of Attack (Continuous) (b) Angle of Attack (Discrete)

Fig. 2: Angle of attack response under LQR control.



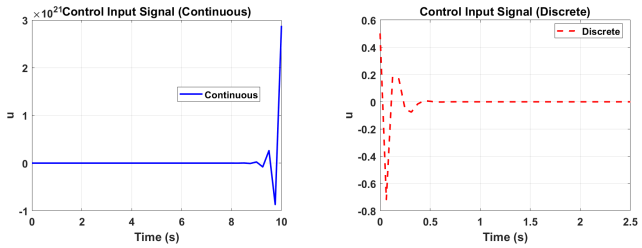
(a) Pitch Angle (Continuous) (b) Pitch Angle (Discrete)

Fig. 3: Pitch angle response under LQR control.



(a) Pitch Rate (Continuous) (b) Pitch Rate (Discrete)

Fig. 4: Pitch rate response under LQR control.



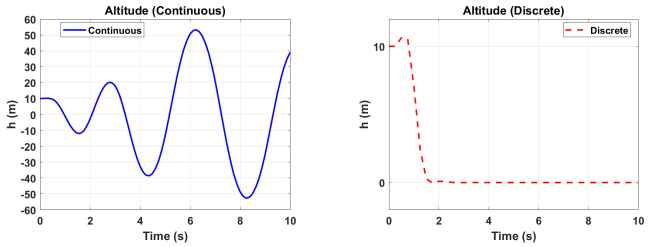
(a) Control Input (Continuous) (b) Control Input (Discrete)

Fig. 5: Control signal under LQR control.

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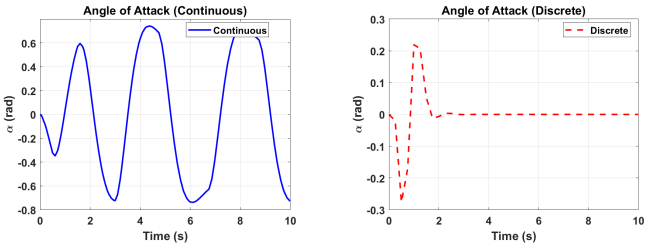
C. LQR with Input Saturation

In this section, the performance of continuous- and discrete-time LQR controllers under actuator saturation is compared. As shown in the altitude (h), angle of attack (α), pitch angle (θ) and pitch rate (q) plots, the discrete-time controller's input signal rapidly settles to zero once it reaches the saturation limit, thereby stabilizing the system, whereas the continuous-time controller's signal continues to oscillate within the bounds, exhibiting an unstable behavior.



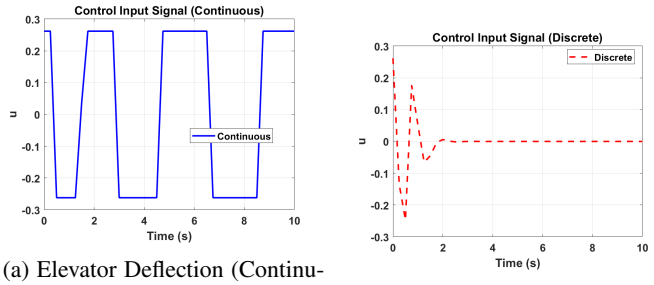
(a) Altitude (Continuous) (b) Altitude (Discrete)

Fig. 6: Altitude response under LQR with input saturation.



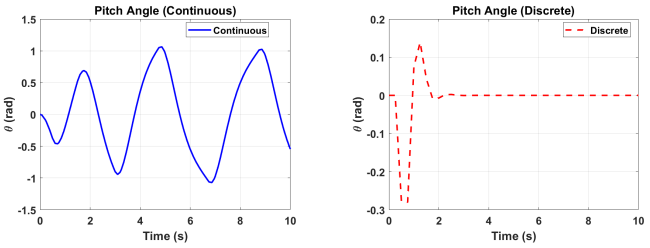
(a) Angle of Attack (Continuous) (b) Angle of Attack (Discrete)

Fig. 7: Angle of attack response under LQR with input saturation.



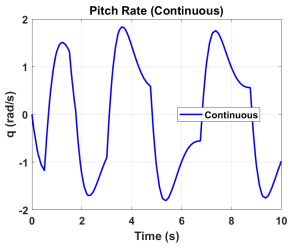
(a) Elevator Deflection (Continuous) (b) Elevator Deflection (Discrete)

Fig. 8: Elevator deflection under LQR control with saturation.

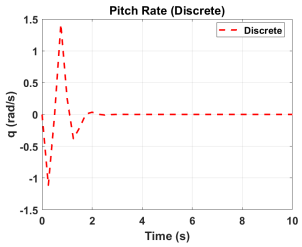


(a) Pitch Angle (Continuous) (b) Pitch Angle (Discrete)

Fig. 9: Pitch angle response under LQR with input saturation.



(a) Pitch Rate (Continuous)



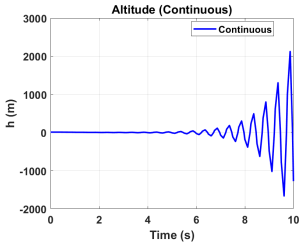
(b) Pitch Rate (Discrete)

Fig. 10: Pitch rate response under LQR with input saturation.

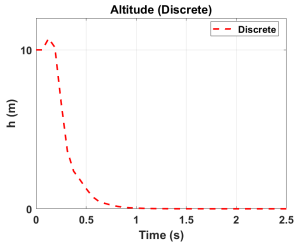
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D. Unconstrained MPC and Horizon Selection

Below are the responses of the continuous-time and discrete-time systems under an unconstrained MPC controller for various states. These plots will facilitate selecting the smallest prediction horizon N that achieves performance equivalent to the LQR benchmark.

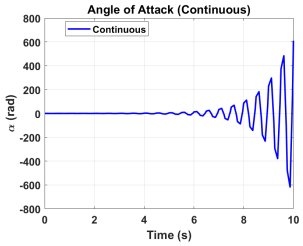


(a) Altitude (Continuous)

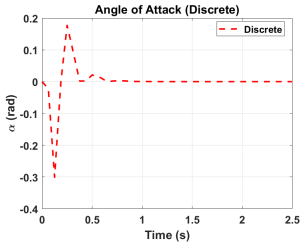


(b) Altitude (Discrete)

Fig. 11: Altitude response under unconstrained MPC.

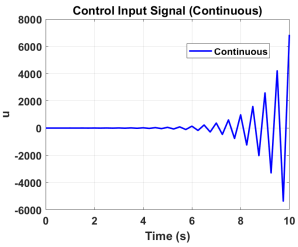


(a) AoA (Continuous)

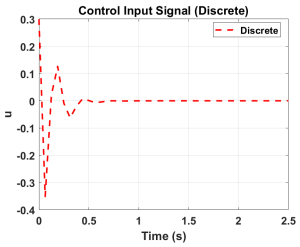


(b) AoA (Discrete)

Fig. 12: Angle of attack response under unconstrained MPC.

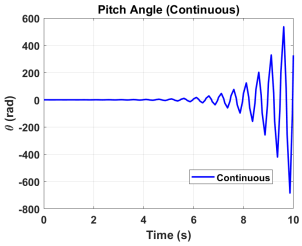


(a) Control Input (Continuous)

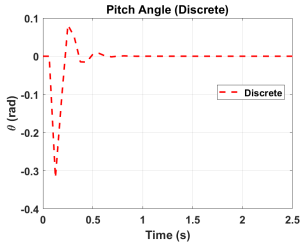


(b) Control Input (Discrete)

Fig. 13: Elevator deflection under unconstrained MPC.

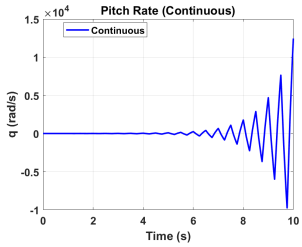


(a) Pitch Angle (Continuous)

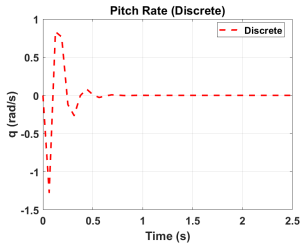


(b) Pitch Angle (Discrete)

Fig. 14: Pitch angle response under unconstrained MPC.



(a) Pitch Rate (Continuous)



(b) Pitch Rate (Discrete)

Fig. 15: Pitch rate response under unconstrained MPC.

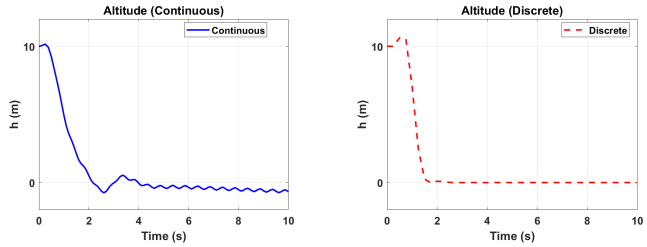
The unconstrained MPC with the chosen horizon $N = 5$ closely matches the LQR benchmark in steady-state behavior while exhibiting a slightly slower rise time in the continuous-time plots. The discrete-time responses settle rapidly with minimal overshoot across all states. In particular, the altitude and pitch-angle curves demonstrate smooth convergence, the angle of attack remains within safe bounds throughout the maneuver, and the elevator deflection (control input) is free of chattering. Overall, these results confirm that the selected horizon achieves performance comparable to LQR without imposing input constraints.

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E. MPC with Input Constraints (N=10)

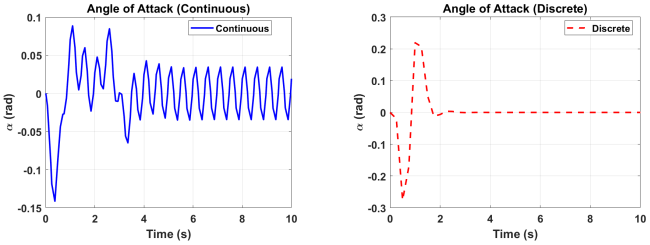
Under an elevator deflection constraint of ± 0.262 rad and with a prediction horizon of $N = 10$, the continuous-time MPC control signal exhibits severe oscillations upon saturation, repeatedly toggling between the bounds, whereas the discrete-time signal rapidly settles to zero without any chattering once the constraint is reached. This behavior is reflected in the

altitude response: continuous oscillations persist beyond 3 s while the discrete system stabilizes around 2 s. In the angle-of-attack plot, the continuous trace shows periodic oscillations of approximately ± 0.05 rad, whereas the discrete response transitions smoothly and remains close to zero. Similarly, the pitch angle and pitch rate mirror the control signal oscillations in the continuous case—yielding sustained ± 0.1 rad and ± 0.5 rad/s oscillations—while the discrete design damps out within roughly 1 s. These findings illustrate that post-saturation chattering in the continuous-time MPC induces periodic disturbances in the plant, whereas the discrete-time MPC achieves rapid and stable equilibrium.



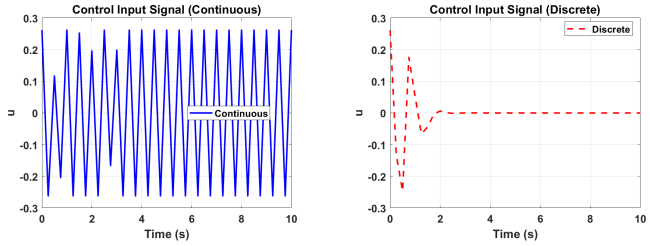
(a) Altitude (Continuous) (b) Altitude (Discrete)

Fig. 16: Altitude response under constrained MPC ($N = 10$).



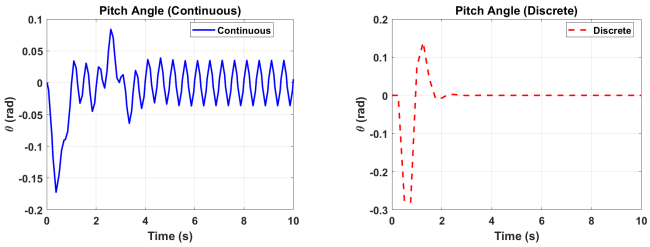
(a) AoA (Continuous) (b) AoA (Discrete)

Fig. 17: Angle-of-Attack response under constrained MPC ($N = 10$).



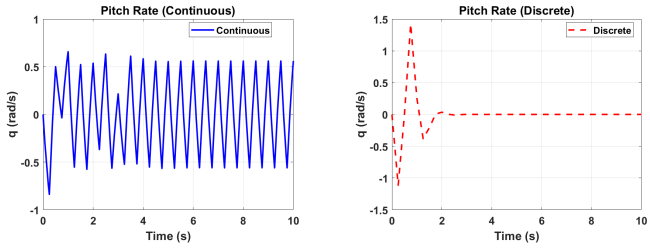
(a) Control Input (Continuous) (b) Control Input (Discrete)

Fig. 18: Control-signal response under constrained MPC ($N = 10$).



(a) Pitch Angle (Continuous) (b) Pitch Angle (Discrete)

Fig. 19: Pitch-angle response under constrained MPC ($N = 10$).



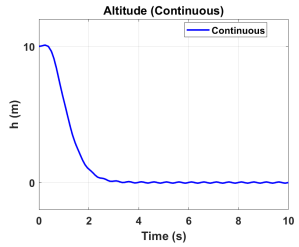
(a) Pitch Rate (Continuous) (b) Pitch Rate (Discrete)

Fig. 20: Pitch-rate response under constrained MPC ($N = 10$).

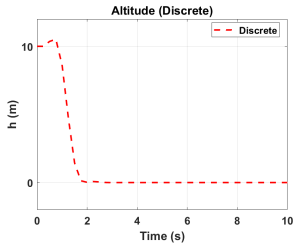
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F. MPC with Input Rate Constraint

An MPC controller with both an elevator deflection constraint of ± 0.262 rad and a rate-of-change constraint $|u[k+1] - u[k]| \leq 0.524\,T_s$ was implemented for $N = 10$. The simulation results indicate that chattering and abrupt command changes are largely suppressed in both continuous- and discrete-time implementations. In the altitude response, the discrete-time MPC settles by approximately 2 s, whereas the continuous-time solution exhibits mild periodic oscillations until about 3 s. In the angle of attack, the continuous trace shows damped oscillations of roughly ± 0.05 rad, while the discrete response decays more rapidly. The control input in the continuous MPC produces regular rate oscillations around ± 0.1 rad, whereas the discrete MPC applies the rate constraint once and then maintains an almost constant command, yielding a much smoother profile. Similarly, the pitch angle and pitch rate in the continuous model display sustained periodic fluctuations, whereas the discrete model damps out within about 1 s and reaches the reference equilibrium. These results demonstrate that the addition of an input rate constraint effectively reduces abrupt control changes and oscillations, enhancing MPC performance, with the discrete-time design achieving both rapid convergence and stable closed-loop behavior.

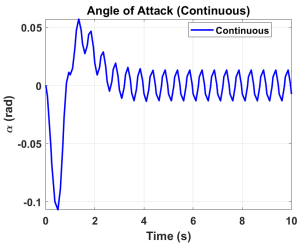


(a) Altitude (Continuous)

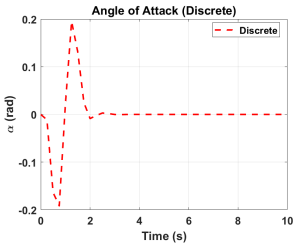


(b) Altitude (Discrete)

Fig. 21: Altitude response under MPC with input-rate constraint ($N = 10$).

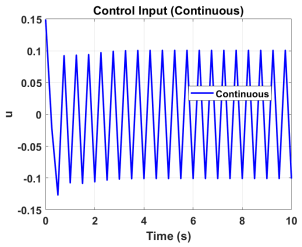


(a) AoA (Continuous)

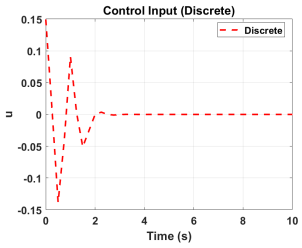


(b) AoA (Discrete)

Fig. 22: Angle-of-Attack response under MPC with input-rate constraint ($N = 10$).

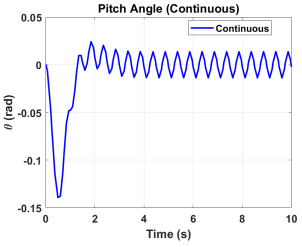


(a) Elevator Deflection (Continuous)

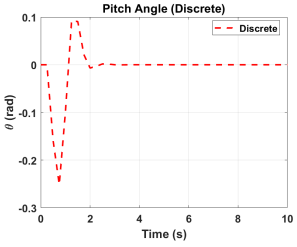


(b) Elevator Deflection (Discrete)

Fig. 23: Elevator deflection under MPC with input-rate constraint ($N = 10$).

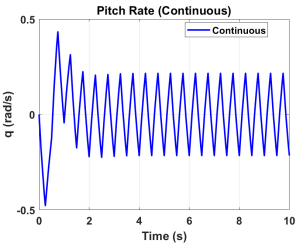


(a) Pitch Angle (Continuous)

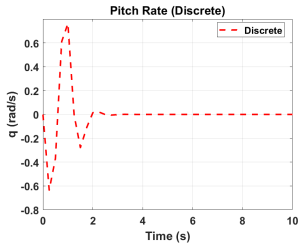


(b) Pitch Angle (Discrete)

Fig. 24: Pitch-angle response under MPC with input-rate constraint ($N = 10$).



(a) Pitch Rate (Continuous)



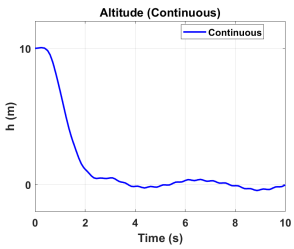
(b) Pitch Rate (Discrete)

Fig. 25: Pitch-rate response under MPC with input-rate constraint ($N = 10$).

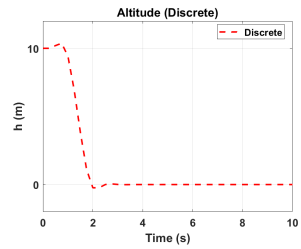
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G. MPC with State Constraint (Pitch Rate)

In MPC simulations with a state constraint on pitch rate ($|q[k]| \leq q_{\max}$) and a horizon of $N = 10$, the continuous-time control signal exhibits periodic oscillations that propagate into the altitude and angular responses, causing mild fluctuations beyond 3 s despite no input constraints. In contrast, the discrete-time MPC meets the pitch-rate constraint and then locks its control input, fully damping oscillations by approximately 2 s and achieving rapid stabilization. The angle of attack and pitch angle in the continuous model display prolonged damped oscillations around ± 0.05 rad, whereas the discrete model suppresses these oscillations within about 1 s and closely tracks the reference. These results demonstrate that, even with only a pitch-rate state constraint, the discrete-time MPC design more effectively attenuates control signal oscillations and output disturbances compared to the continuous-time implementation.

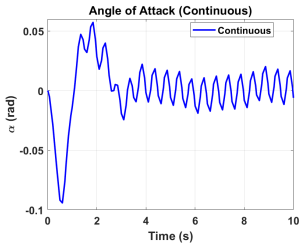


(a) Altitude (Continuous)

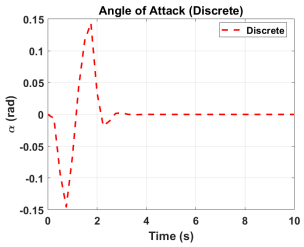


(b) Altitude (Discrete)

Fig. 26: Altitude response under MPC with state constraint ($N = 10$).

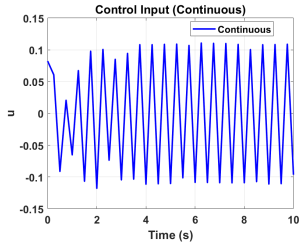


(a) AoA (Continuous)

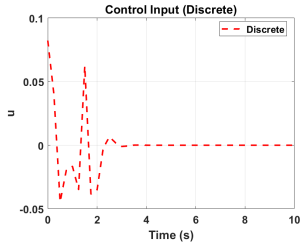


(b) AoA (Discrete)

Fig. 27: Angle-of-attack response under MPC with state constraint ($N = 10$).

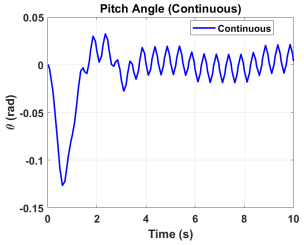


(a) Control Input (Continuous)

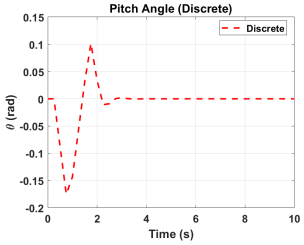


(b) Control Input (Discrete)

Fig. 28: Control input response under MPC with state constraint ($N = 10$).

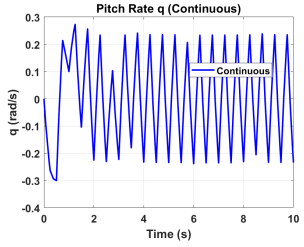


(a) Pitch Angle (Continuous)

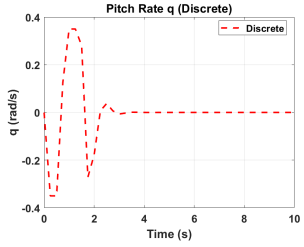


(b) Pitch Angle (Discrete)

Fig. 29: Pitch angle response under MPC with state constraint ($N = 10$).



(a) Pitch Rate (Continuous)



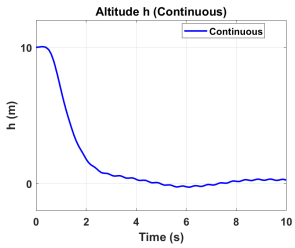
(b) Pitch Rate (Discrete)

Fig. 30: Pitch rate response under MPC with state constraint ($N = 10$).

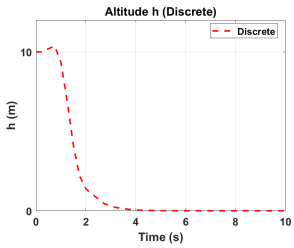
*

H. MPC with Horizon Selection

By iteratively reducing the prediction horizon, the smallest horizon length that stabilizes the system under option g was found to be $N = 5$. Below are the continuous-time and discrete-time responses obtained when enforcing an input constraint under MPC with a prediction horizon of $N = 5$.

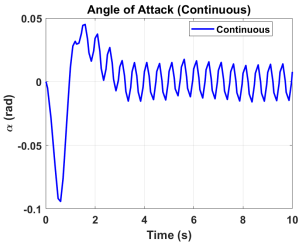


(a) Altitude (Continuous)

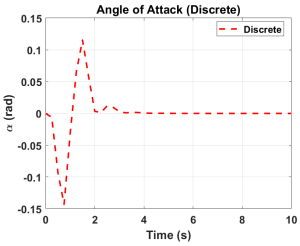


(b) Altitude (Discrete)

Fig. 31: Altitude response under MPC with altitude state constraint ($N = 10$).

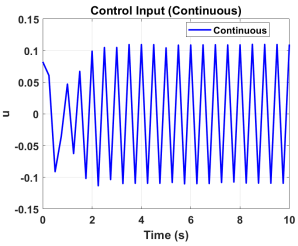


(a) AoA (Continuous)

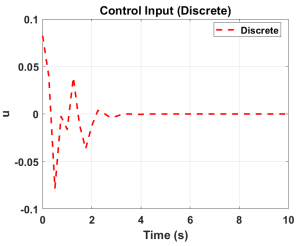


(b) AoA (Discrete)

Fig. 32: Angle-of-attack response under MPC with altitude state constraint ($N = 10$).

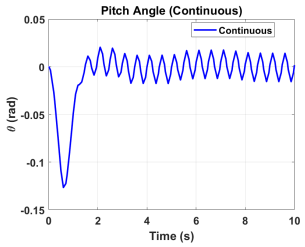


(a) Control Input (Continuous)

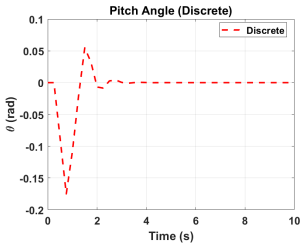


(b) Control Input (Discrete)

Fig. 33: Control-signal response under MPC with altitude state constraint ($N = 10$).

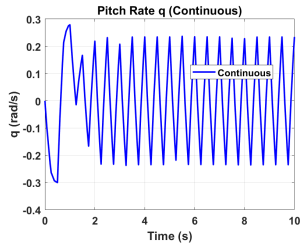


(a) Pitch Angle (Continuous)

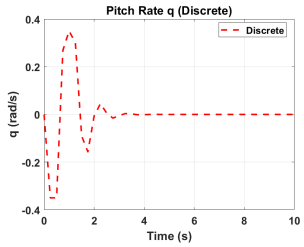


(b) Pitch Angle (Discrete)

Fig. 34: Pitch-angle response under MPC with altitude state constraint ($N = 10$).



(a) Pitch Rate (Continuous)



(b) Pitch Rate (Discrete)

Fig. 35: Pitch-rate response under MPC with altitude state constraint ($N = 10$).

III. Results

Comparing the developed control strategies, the discrete-time designs consistently achieve faster and more stable convergence than their continuous-time counterparts. When actuator saturation is imposed on the LQR controller, the discrete-time LQR signal immediately settles upon reaching the limits—rapidly suppressing oscillations—whereas the continuous-time LQR generates prolonged, high-amplitude oscillations after saturation.

In unconstrained MPC, by iteratively reducing the prediction horizon, the minimum stable horizon was found to be $N = 5$; at this setting, the MPC responses closely replicate the ideal LQR behavior in both amplitude and damping. Introducing both the input limit and the rate-of-change constraint $|u[k+1] - u[k]| \leq 0.524\,T_s$ significantly reduces chattering and abrupt command changes in both designs, yet the discrete-time MPC still delivers a smoother, faster transition—reaching equilibrium within 1–2 s.

Finally, even with only a pitch-rate state constraint, the discrete-time MPC design attenuates periodic oscillations far more effectively than the continuous-time implementation, confirming the superior disturbance rejection and stabilization performance of the discrete-time approach.

IV. Refernces

References

[1] Ugur Yildiran Lecture Notes on Model Predictive Control, Department of Control and Automation Engineering, Spring 2025.